

Simulation Study 1

STUDIES ON RECTANGULAR WAVEGUIDE

Objective:

- Study standing wave distribution inside a waveguide with short-circuit termination.
- Calculation of guide wavelength λ_g and hence plot of dispersion diagram (ω - β plot).

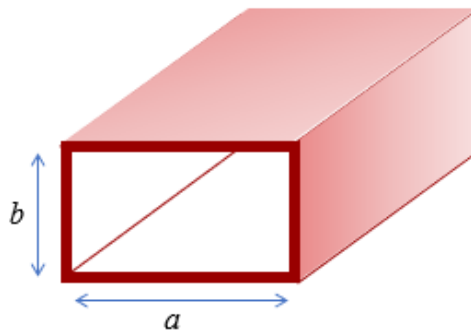


Fig.1 A rectangular waveguide.

A waveguide is a hollow metal pipe used to carry high power microwave signals. It is widely used in microwave ovens, different types of radars, satellite communications, and microwave wireless radio links. Among different shapes, rectangular waveguide is the most popular guide. Its dimension is so selected that only a single mode (usually the lowest order TE_{10} mode) is supported with least attenuation over a given frequency band. For the mode, a waveguide behaves as a high pass filter. The cut-off frequency of an air-filled waveguide is $f_c = c/(2a)$, where c is the velocity of light in air and a is the broadside internal dimension of the waveguide. Attenuation of the propagating wave down the guide is a function of frequency. Attenuation is high near the cut-off frequency. The only source of loss inside an air-filled (dry air, water vapour is a lossy dielectric) waveguide is the conductor loss. When the guide is filled by a dielectric material of relative permittivity ϵ_r , the cut-off frequency decreases by a factor of $\sqrt{\epsilon_r}$. It also introduces dielectric loss, which depends on the loss tangent of the dielectric material.

If the other end of the waveguide is kept open, it will radiate a fraction of the electromagnetic wave from the open-end i.e. the guided wave transforms into space wave. Therefore, an open-ended waveguide can be considered as an antenna. On the other hand, if the end is terminated with a short circuit (a brass plate), the forwarded traveling wave is reflected back from the short plane. These two waves propagating in opposite directions create standing wave pattern inside the waveguide, as shown in Fig. 2. The separation between two successive minima or two successive maxima is half guide wavelength i.e. $\lambda_g/2$.

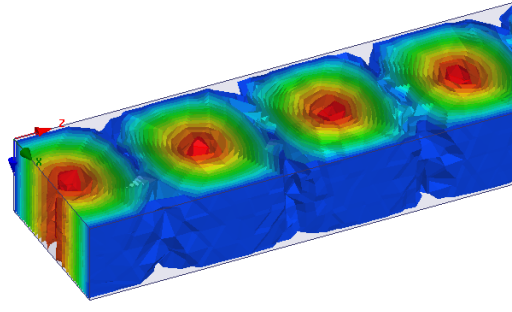


Fig. 2 Time averaged scalar electric field distribution inside a waveguide for TE₁₀ standing wave pattern.

Simulation study:

Draw a section of an air-filled WR-90 waveguide (inner dimensions: $a = 22.86$ mm, $b = 10.16$ mm) of length 30 cm. Consider thickness of the brass wall as 1.0 mm.

Variation of the guided wavelength λ_g and plot of dispersion diagram:

- Connect Wave Port to one end and terminate the other end by short-circuit (a metal plate).
- Simulate the structure over 8.2-12.4 GHz (suggested analysis frequency in HFSS is 10.0 GHz, frequency resolution at least 0.1 GHz, set analysis set up as “fast”).
- Plot vector electric field and magnetic field inside the guide for the TE₁₀ mode.
- Note down λ_g at 10-12 frequency points.
- Compare the simulated and calculated λ_g in a table from the expression given below.

$$\lambda_g = \frac{\lambda_o}{\left[1 - \left(\frac{\lambda_o}{2a}\right)^2\right]^{1/2}}, \quad (4)$$

where λ_o is free space wavelength and a is the broad internal dimension of the guide.

- Plot the dispersion diagram (β/k_0 vs. f , where $k_0 = 2\pi/\lambda_o$) over 8.2-12.4 GHz. For a lossless line the phase constant β is

$$\beta = \frac{2\pi}{\lambda_g}. \quad (5)$$

- Simulate the waveguide over 13-18 GHz and plot vector electric field distribution inside the guide for the TE₂₀ mode.

Simulation Study 2

DESIGN OF PASSIVE COMPONENTS BASED ON MICROSTRIP LINE

Objective:

- Study electric and magnetic field distributions on a cross-sectional plane of a microstrip line.
- Calculation of guide wavelength λ_g and hence plot of dispersion diagram.
- Study different sources of loss.
- Design a quarter-wavelength impedance transformer.
- Design a single stub matching section.
- Design a patch antenna.

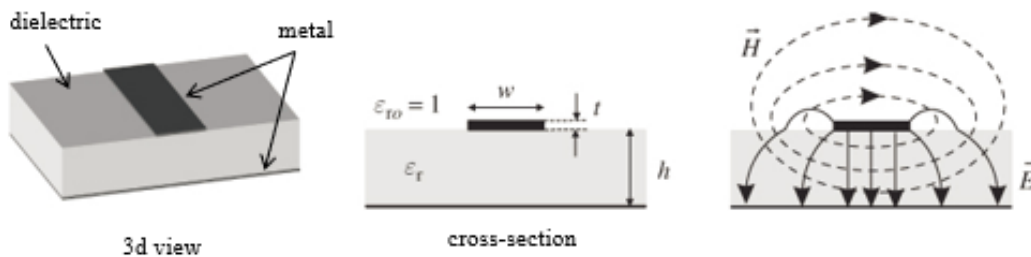


Fig. 1 A microstrip line and electric and magnetic fields on a cross sectional plane.

Microstrip line is one of the most popular planar guiding structures. It is fabricated using printed circuit board technology. As shown in Fig. 1, it consists of a conducting strip separated from a ground plane by a dielectric layer known as the substrate. Microwave components such as antennas, couplers, filters, power dividers etc. can be formed from microstrip with the entire device existing as the pattern of metallization on the substrate. It also allows convenient integration of passive and active components. Many types of microwave circuits and subsystems are made in microstrip form in a single printed circuit board. It is less expensive than traditional waveguide technology as well as being far lighter and more compact. However, compared to air-filled waveguide, loss increases, and power handling capability decreases.

The microstrip line is always used in TEM mode excitation. Thus, it does not have any cutoff frequency and can be described by a characteristic impedance Z_0 . Characteristic impedance Z_0 of a microstrip line mainly depends on the thickness of the substrate h and dielectric constant of the substrate ϵ_r . The substrate parameters h , ϵ_r and metal thickness t is defined by the manufacturer. Typical strip thicknesses are $17\text{ }\mu\text{m}$, $35\text{ }\mu\text{m}$ etc. Note that t should be at least five times the skin depth of the metal (usually copper) at the lowest frequency of operation. A closed form expression to find the dimensions for a given characteristic impedance Z_0 is

$$\begin{aligned} \frac{W}{d} &= \frac{8e^A}{e^{2A} - 2} && \text{for } W/d < 2, \\ &= \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left\{ \ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right\} \right] && \text{for } W/d < 2, \end{aligned} \quad (1)$$

where

$$\begin{aligned} A &= \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right), \\ B &= \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}. \end{aligned}$$

The microstrip line is associated with metal losses, dielectric loss, radiation loss and loss due to power lost into surface waves. All the loss components increase with frequency. Dielectric loss depends on the loss tangent ($\tan\delta$) of the substrate material. However, a lossless transmission line model is frequently used to calculate the characteristic impedances and electrical lengths of a microwave component based on microstrip line. The transmission line model cannot predict accurate insertion loss or reflection loss. A full wave electromagnetic software is used for that purpose.

Group delay:

Group delay is a measure of phase distortion. It is the actual transit time of a signal through a device under test and is a function of frequency. Group delay is measured as

$$\tau_d = -\frac{\partial\theta}{\partial\omega} \approx -\frac{1}{2\pi} \frac{\Delta\angle S_{21}}{\Delta f}. \quad (2)$$

Quarter-wave transformer: single frequency matching for a resistive load

The lossless transmission line model can be used to calculate the electrical parameters of a quarter-wave transformer. Then, the closed form expressions are used to obtain the initial physical dimensions of the matching section. In Fig. 2, the input impedance Z_{in} should be Z_0 to make the reflection coefficient $\Gamma = 0$. The condition of matching is

$$Z_1 = \sqrt{Z_0 R_L}, \quad \theta = 90^\circ @ f_0. \quad (3)$$

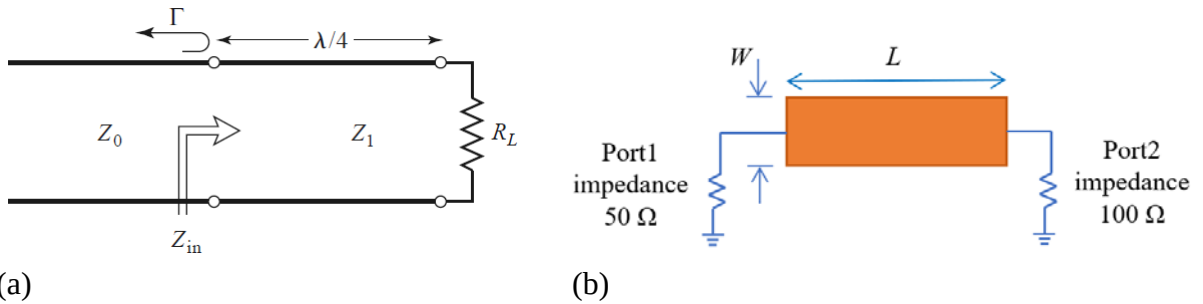


Fig. 2 The quarter-wave transformer, (a) lossless transmission line model and (b) microstrip line implementation, physical length corresponds to equivalent electrical length of $90^\circ (= \lambda_g/4)$ at the design frequency f_0 .

Single stub matching: single frequency matching for any complex load

A proper length of an open or shorted transmission line section can provide any desired value of reactance or susceptance. For a given susceptance or reactance, the difference in lengths of an open- or short-circuited stub is $\lambda_g/4$. The basic idea is to select an appropriate d in Fig. 3(a) so that the admittance Y , looking into the line at distance d from the load, is of the form $Y_0 + jB$, Y_0 being $1/Z_0$. Then the stub susceptance is chosen as $-jB$, resulting in a matched

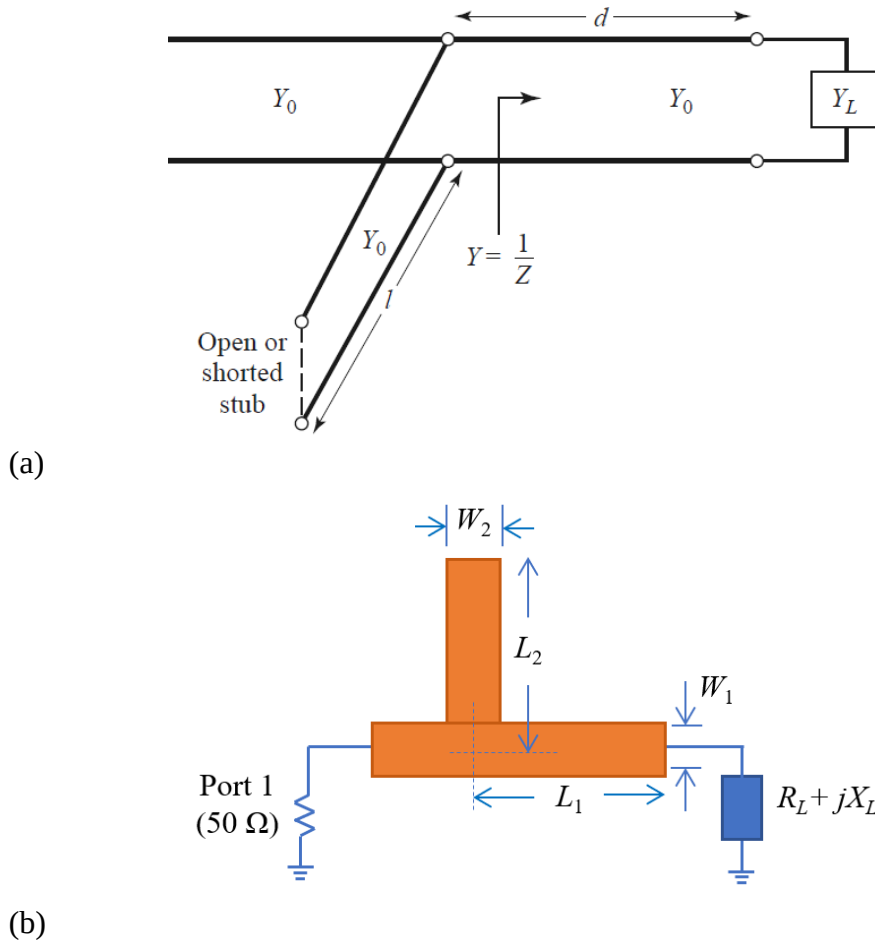


Fig. 3 The single stub matching section, (a) lossless transmission line model and (b) microstrip line layout.

condition. (Refer to Smith chart and analytic solutions in D. M. Pozar's Microwave Engineering, chapter 5 for detail procedure).

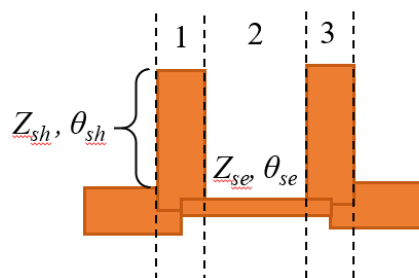


Fig. 4 Different sections of the BSF.

Bandstop filter (BSF):

A U-section stub behaves as a BSF with maximum rejection at f_0 when the series and shunt arm lengths are 90° . A straightforward method to obtain the complex S-parameters of the BSF shown in Fig. 4 is by calculating overall ABCD matrix. The BSF is first divided into three regions. Then, the overall ABCD matrix is

$$[ABCD] = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_{se} & jZ_{se} \sin \theta_{se} \\ jY_{se} \sin \theta_{se} & \cos \theta_{se} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}, \quad (4)$$

where the shunt open-stubs are represented by their equivalent input susceptances Y , and the connecting section as a section of transmission line. The characteristic impedance Z_{sh} should be as low as possible. Similarly, Z_{se} should be as high as possible. Their values are limited by fabrication limit, higher order mode excitation, and loss. The electrical lengths are 90° at the design frequency f_0 . Then, at any a frequency f ,

$$\theta_{sh,se} = \frac{\pi}{2} \left(\frac{f}{f_0} \right). \quad (5)$$

The corresponding S-parameters are

$$S_{11} = \frac{A + B/Z_0 - C Z_0 - D}{A + B/Z_0 + C Z_0 + D}, \quad \text{and} \quad S_{21} = \frac{2}{A + B/Z_0 + C Z_0 + D}. \quad (6)$$

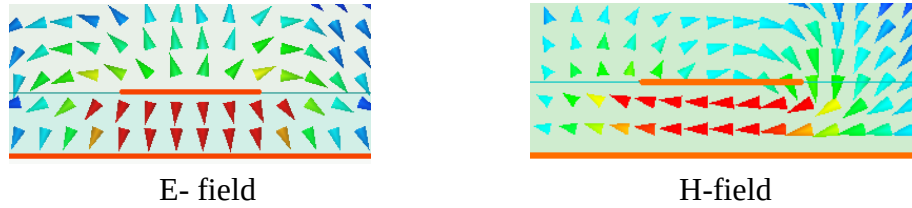


Fig. 5 Vector field distributions on a cross-sectional plane.

Simulation study:

A. Microstrip line

Draw a section of a microstrip line ($L = 50$ mm, say) of characteristic impedance 50Ω on the substrate as specified. Consider the port impedances as 50Ω .

- Simulate the structure over 8-12 GHz and plot $|S_{11}|$, $|S_{21}|$ in dB and $\angle(S_{21})$.
- Plot total loss $(1 - |S_{11}|^2 - |S_{21}|^2)$.
- Calculate and plot individual component of losses: (i) metal loss, (ii) dielectric loss and (iii) radiation + surface wave loss (Hints: Metal loss is zero if copper is replaced by PEC. Dielectric loss is zero if $\tan \delta = 0$).
- Calculate and plot β/k_0 vs. f .
- Calculate and plot τ_d vs. f .
- Draw the vector electric and magnetic field distributions on a cross-sectional plane at a frequency 10 GHz, say, as shown in Fig. 5.

B. Quarter-wave transformer:

Design a quarter-wave transformer to match a $100\ \Omega$ impedance at a given frequency f_0 .

- Implement it using a microstrip line as shown in Fig. 2.
- Calculate and plot $|S_{11}|$ dB and $|S_{21}|$ dB in MATLAB over a frequency range $0.8 f_0$ to $1.2 f_0$.
- Plot the simulated $|S_{11}|$ dB and $|S_{21}|$ dB and compare with the calculated response.

C. Single stub matching:

- As shown in Fig. 3, design a single stub matching section for a specified impedance $R_L + jX_L$ at a given frequency f_0 .
- Obtain overall ABCD parameters of the matching section.
- Obtain the corresponding $|S_{11}|$ using (6).
- Plot $|S_{11}|$ dB in MATLAB over a frequency range $0.8 f_0$ to $1.2 f_0$.
- Implement the matching section in the full wave simulator using the specified substrate.
- Fine tune for minimum reflection at f_0 .
- Plot the simulated $|S_{11}|$ dB and compare with the MATLAB calculated $|S_{11}|$ dB.

D. Bandstop filter (BSF):

- Considering $50\ \Omega$ port termination, choose the parameters of the BSF as suggested above.
- Calculate and plot $|S_{11}|$ dB and $|S_{21}|$ dB in MATLAB over a frequency range $0.8 f_0$ to $1.2 f_0$.
- Implement the BSF in the full wave simulator using the specified substrate.
- Fine tune the physical parameters to place the transmission zero at the specified f_0 .
- Plot the simulated $|S_{11}|$ dB, $|S_{21}|$ dB and compare with the calculated responses.

Simulation Study 3

DESIGN OF A PATCH ANTENNA

Objective:

- Design of a microstrip patch antenna with microstrip feed.
- Study the antenna characteristics.

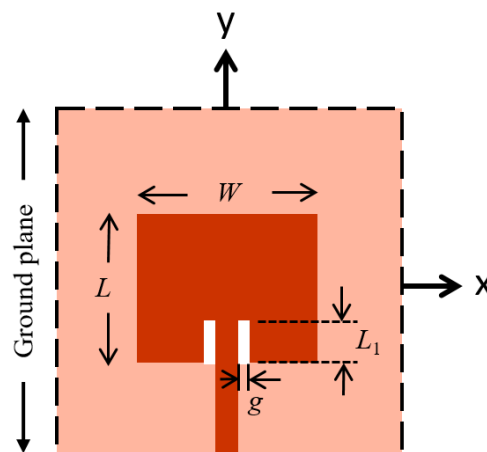


Fig.1 A rectangular patch antenna.

A patch antenna consists of a flat rectangular patch of metal, mounted over a dielectric slab backed by sheet of metal called a ground plane. Two metal sheets together form a leaky resonator, the length L being approximately one-half wavelength at the radiating frequency. The patch antenna is mainly practical at microwave frequencies. A microstrip line of characteristic impedance $50\ \Omega$ is used to excite the antenna. Several other excitation schemes exist. Then, the antenna radiates from the discontinuities at feed and opposite edges. Multiple patch antennas on the same substrate can be used to make high gain array antennas, and phased arrays in which the beam can be electronically steered.

An antenna is characterized by the input reflection coefficient (depends on input impedance matching) and radiation pattern over its operating frequencies. Radiation pattern or antenna pattern or far-field pattern refers to the directional dependence of the strength of the radio waves transmitted by the antenna. The far-field pattern is usually represented graphically as a plot of the electric field strength or power (square of electric field) at a constant radius. The plot is typically represented as a three-dimensional graph in a spherical coordinate system (see, Fig. 2), or as separate graphs in the vertical plane (or elevation plane showing variation with θ) and horizontal plane (or azimuth plane showing variation with ϕ). The antenna shown in Fig. 1 radiates maximum power in $+Z$ direction. The radiated electric field is parallel to yz -plane. Then, this plane is called the E-plane of the antenna. The zx -plane is known as H-plane.

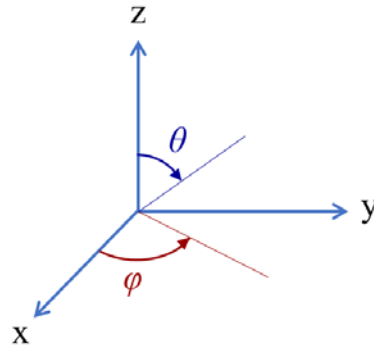


Fig. 2 Spherical co-ordinate system.

Antenna parameters:

Far-field range:

The far-field range of an antenna is defined by the range where the radiated field components dominates over the bounded field components. The radiated field propagates through space and its electric field and magnetic field are in the same phase. The bounded field component cannot propagate through space and its electric and magnetic fields are in phase quadrature. This field component has similar characteristics as those inside an air-filled coil. The far-field range is defined by the approximate relationship

$$r \geq \frac{2D^2}{\lambda_0}, \quad (1)$$

where D is the maximum dimension of the antenna and λ_0 is the free space wavelength of the radio wave. Go through any antenna book for detail.

Antenna efficiency:

Efficiency of an antenna is defined as the ratio of the power that is radiated to the total power used by the antenna. A part of the incident power is lost as heat. Thus, efficiency of a practical antenna is always less than 100%. Mathematically, efficiency is equal to the radiation resistance divided by total resistance (real part) of the feed-point impedance.

Directivity:

Directivity of an antenna is the ratio of maximum radiation intensity (power per unit surface) radiated by the antenna in the maximum direction divided by the intensity radiated by a hypothetical isotropic antenna radiating the same total power as that antenna. For example, a hypothetical antenna which has a radiation pattern of a hemisphere would have a directivity of 2. Directivity is a dimensionless ratio and may be expressed numerically or in decibels. Directivity is identical to the peak value of the directive gain. It is specified without respect to antenna efficiency thus differing from the gain.

Gain:

The gain or power gain of an antenna is defined as the ratio of the intensity (power per unit surface) actually radiated by the antenna in a given direction at an arbitrary distance divided by the intensity radiated at the same distance by a hypothetical isotropic antenna. The fraction

of power lost in the antenna structure is included in the gain. Thus, maximum gain = $k \times$ directivity, where k is the antenna efficiency.

Friis transmission equation:

An isotropic antenna radiates power uniformly in all direction. If the transmitted power is P_t , then power passing through unit area at a distance r is $P_t/(4\pi r^2)$. Now, if the transmitting antenna has a gain of G_t and a receiving antenna with a gain G_r is placed at a distance r , received power is

$$P_r = P_t G_t G_r \left(\frac{\lambda_0}{4\pi r} \right)^2, \quad (2)$$

where λ_0 is the free space wavelength of the electromagnetic signal. In decibel scale,

$$10 \log_{10} P_r = 10 \log_{10} P_t + 10 \log_{10} G_t + 10 \log_{10} G_r + 20 \log_{10} \left(\frac{\lambda}{4\pi r} \right) \text{ dB}. \quad (3)$$

The above relationship assumes that $r \gg \lambda_0$, the antennas are unobstructed, no multipath, no atmospheric scattering or attenuation effect, bandwidth is narrow enough for single frequency consideration (no pulse), antennas are ideal with no feed line loss and are in right polarization.

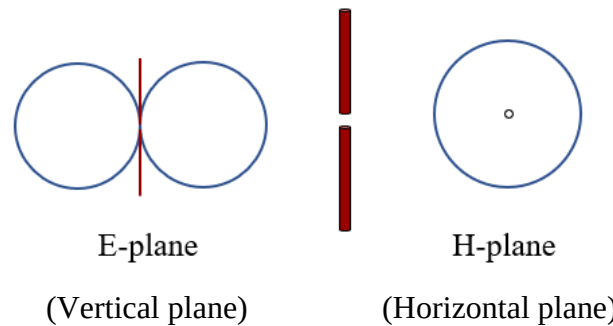


Fig. 3 E- and H-plane patterns of a half-wave dipole antenna.

Polarization:

The electric field determines the polarization or orientation of the radio wave. A linearly polarized antenna radiates wholly in one plane containing the direction of propagation. A linearly polarized antenna is said to be vertically polarized when its electric field is perpendicular to the Earth's surface. An example of a vertical antenna is the "whip" antenna on an automobile. Horizontally polarized antennas have their electric field parallel to the Earth's surface. The radiated electric field may rotate with time (like Lissajous figure in CRO). It is called a circularly polarized antenna when the plane of polarization rotates in a circle making one complete revolution during one period of the wave. If the rotation is clockwise looking in the direction of propagation, the sense is called right-hand-circular (RHC). If the rotation is counter clockwise, the sense is called left-hand-circular (LHC). A circular polarized wave radiates energy in both the horizontal and vertical planes and all planes in between. In a wireless link, both the transmitting and receiving antennas might have similar polarization to maximize the received power.

E-plane and H-plane:

The E-plane and H-plane are reference planes for linearly polarized antennas. E-plane is the plane containing the electric field vector and the direction of maximum radiation. The electric field plane determines the polarization or orientation of the radio wave. For a vertically polarized antenna, the E-plane usually coincides with the vertical plane. Similarly, for a linearly polarized antenna, H-plane is the plane containing the magnetic field vector and the direction of maximum radiation. The magnetic field or H-plane lies at a right angle to the E-plane.

Co-polarization and cross-polarization patterns:

Co-pol is defined as the polarization of the antenna was meant to radiate. Cross-pol is defined as its orthogonal pair. For example, the antenna in Fig. 1 is expected to radiate E-field parallel to yz-plane. However, in practice, it radiates an undesired small E-field component perpendicular to yz-plane. Then, the radiation pattern showing this undesired component is called the cross-pol pattern of the antenna. A purely polarized antenna will have low cross polarized radiation. A measure of how purely polarized an antenna is, is the cross polarization level. It is defined as the difference in decibels between the maximum radiation intensity of the co and cross polarizations, respectively.

Beamwidth:

The half power beamwidth or 3 dB beamwidth of an antenna is the angle between the half-power (-3 dB) points of the main lobe. They may have different value in the E- and H-planes. Similarly, another frequently used parameter is first null-to-null beamwidth.

Impedance bandwidth:

A fraction of the power fed to an antenna reflects back from the feed end towards the source. The reflection coefficient depends on the mismatch between the feed line and the antenna input impedance. For small antennas, impedance bandwidth is defined as the frequency band over which reflection coefficient $|S_{11}| \leq -10$ dB. The definition of bandwidth may vary from application to applications.

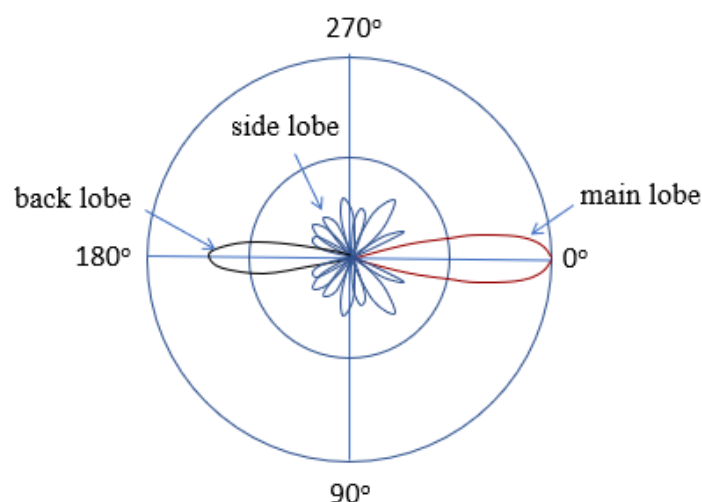


Fig. 4 Two-dimensional radiation pattern of an antenna.

Simulation study:

A. Design of a microstrip patch antenna:

As shown in Fig. 1(a), calculate the length of the antenna $L \approx \lambda_g/2$ for the specified operating frequency f_0 (you can use the online patch calculator). In the first step, consider the initial values as $W = 1.25L$, $g = 1$ mm, $L_1 = L/3$. The characteristic impedance of the feed line is 50Ω . The ground plane of the antenna should be at least $3L \times 3L$. Plot the delay of S_{11} . The highest delay is at the resonant frequency of the patch. If the resonant frequency is f_1 then to transform it to f_0 , multiply L by f_1/f_0 . Optimize L_1 for best input matching (target is $|S_{11}| < -25$ dB at f_0). You may have to multiply the length a second time to change the resonant frequency to f_0 . Show all the antenna dimension in your drawing.

Calculation:

1. Plot return loss and calculate the 10 dB matching bandwidth.
2. Plot the three-dimensional realized gain pattern (understand different type of gains of an antenna). Find the peak realized gain value.
3. Plot realized gain pattern of the antenna at f_0 on the two principal planes containing the main beam (identify E-plane and H-plane). Note down the maximum gain and its direction (θ , ϕ), and 3 dB beamwidth.
4. Plot realized co-polarization patterns of the antenna at f_0 on the same principal planes. Also plot the cross-polarization patterns.